

[N.B. – Answer all the questions. Each question carries ONE mark. Block fully, with a black ball- point pen, the circle of the letter that stands for the correct/best answer in the “Answer sheet” for the Multiple Choice Questions Examination.]

Candidates are asked not to leave any mark or spot on the question paper.

1. For a mesokurtik distribution, $\beta_2 = -$
(a) 0 (b) -3 (c) 3 (d) 1
- Answer the next TWO questions based on the following information
- For a frequency distribution, the following central moments are given:
$$\mu_2 = 4, \quad \mu_3 = -4, \quad \mu_4 = 48$$
2. What is the standard deviation of the distribution?
(a) 2 (b) 4 (c) 8 (d) 16
3. What is Karl Pearson’s coefficient of skewness based on moments ($\sqrt{\beta_1}$)?
(a) -1.0 (b) -0.5 (c) 0.5 (d) 1.0
4. The correlation coefficient between two variables lies between:
(a) 0 and 1 (b) -1 and 0 (c) -1 and +1 (d) $-\infty$ and $+\infty$
5. If the correlation coefficient between X and Y is 0.6, the correlation between 2X and 3Y is:
(a) 0.6 (b) 0.2 (c) 1.8 (d) Cannot be deter-
mined
6. If the regression coefficients are 0.8 and 0.5, the correlation coefficient is:
(a) 0.40 (b) 0.63 (c) 0.80 (d) 1.30
7. If $Y = 4X - 8$ and $\bar{Y} = 20$, find $\bar{X}$.
(a) 3 (b) 5 (c) 7 (d) 9
8. The following are values of arithmetic, geometric, and harmonic mean, respectively
i. {10, 12, 14}
ii. {10, 10, 10}
iii. {10, 8, 6}
Which one is possible?
(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
9. The values of mode and median of moderately skewed distribution are 60 and 65. What is value of the mean?
(a) 70 (b) 67.5 (c) 72 (d) 55
10. Which of the following are true about the median?
i. It is the middle value when data is ordered
ii. It is unaffected by extreme outliers
iii. It is always equal to the mean in a symmetrical distribution
Which one is correct?
(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
11. A survey categorizing people by their favorite color is an example of which measurement scale?
(a) Nominal (b) Ordinal (c) Interval (d) Ratio

12. Which is a discrete variable?
- (a) Number of languages spoken by a person

(b) Time taken to complete a race

(c) Length of a road

(d) Volume of water in a tank

13. If $x_1 = 3$, $x_2 = 2$, $x_3 = -6$, and $x_4 = 4$, what is $\sum_{i=1}^4 x_i^2$?
- (a) 45

(b) 65

(c) 85

(d) 89

Answer the next TWO questions based on the following information

The table below shows monthly sales (in thousands of dollars) for two products in four regions.

Product	North	South	East	West
A	12	15	10	8
B	18	22	14	16

14. What is the total sales of Product A across all regions?
- (a) 40

(b) 42

(c) 45

(d) 48
15. What is the total sales of all products across all regions?
- (a) 95

(b) 100

(c) 105

(d) 115
16. Which of the following are true about standard deviation?
- i. It has the same unit as the original data

ii. It is the square root of variance

iii. A larger standard deviation indicates greater dispersion

Which one is correct?

- (a) i and ii

(b) i and iii

(c) ii and iii

(d) i, ii and iii
17. If the standard deviation of a dataset is 4, what is its variance?
- (a) 2

(b) 4

(c) 8

(d) 16
18. If the mean of a dataset is 50 and standard deviation is 10, the coefficient of variation is:
- (a) 10%

(b) 15%

(c) 20%

(d) 25%
19. If $Q_1 = 25$ and $QD = 10$, what is the value of Q_3 ?
- (a) 45

(b) 70

(c) 10

(d) 25
20. Which of the following does not depend on change of origin?
- i. Median

ii. Standard deviation

iii. Regression coefficient

Which one is correct?

- (a) i and ii

(b) i and iii

(c) ii and iii

(d) i, ii and iii